Data-Driven Stock Investment Decision Support

A Reinforcement Learning Approach to
Risk-Sensitive Sequential Portfolio Decisions

Master Thesis

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Introduction

The problem of sequential investment under uncertainty motivates this thesis. Modern reinforcement learning offers tools for such decisions, but most methods collapse the return to its expectation. Distributional approaches instead model the full return distribution, as introduced by Dabney et al. [1] with Implicit Quantile Networks. Separately, epistemic uncertainty over a model's own estimates can be quantified through Evidential Deep Learning [2]. This thesis combines both ideas [1], [2] into a single risk-sensitive decision support system.

Box gallery (visual test)

Definition 0.0.1 (Invariants of unimodular symmetric forms)
Let Z be a finitely-generated free abelian group, and let $q : Z \times Z \to \mathbb{Z}$ be a unimodular form. To it we assign rank, type (parity), and signature.

Notation
In [MH73], unimodular symmetric forms on Z are called inner products on Z.
Forms of type I are odd and forms of type II are even.

Example 0.0.2 (E_8)
Define the form E_8 on $\mathbb{Z}^8$. This form is unimodular, positive-definite and even, so $\mathrm{rank}(E_8) = \mathrm{sign}(E_8) = 8$.

Theorem 0.0.3 (Freedman)
Let q be a unimodular, symmetric form. If q is even, there is exactly one closed, oriented, simply-connected topological four-manifold whose intersection form is q.

Proposition 0.0.4 ($w_2(M)$ is mod-2 reduction of characteristic vector)
Let M be a closed, connected, oriented four-manifold with intersection form q_M.
Then for any characteristic vector $w \in H^2(M, \mathbb{Z})$, $w_2(M) = \mathrm{PD}(w)$ mod 2.

Corollary 0.0.5 (Closed four-manifolds admit Spin^c-structures)
Every closed, connected, oriented four-manifold admits Spin^c-structures.

Lemma 0.0.6 (Elkies)
Let $q : Z \times Z \to \mathbb{Z}$ be a symmetric, unimodular bilinear form. If q is not $\oplus(-1)$ nor $\oplus(+1)$, then there exists a characteristic vector w with $|q(w, w)| < \text{rank}(q)$.

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First of all, I would like to thank the mathematical community whose documents make LaTeX typography worth studying. This section is included to test the visual spacing of front matter pages.

CHAPTER 1

Preliminaries

In this chapter we review the basic language of algebraic topology and differential geometry that is needed to discuss Seiberg-Witten theory and the Donaldson theorem. In the differential realm, we assume familiarity with smooth manifolds, de Rham cohomology, vector bundles, and some Riemannian geometry. In algebraic topology, we assume familiarity with singular and cellular homology.

1.1 Gauge theory

Gauge theory is the heart of Seiberg-Witten theory. The name comes from physics, where gauge fields are fields with symmetries and redundant degrees of freedom. Mathematically, the study of gauge fields is done in terms of principal bundles, connections, curvature, and associated vector bundles.

1.1.1 Principal bundles, connections and curvature

The basic objects of gauge theory are principal bundles.

Definition 1.1.1 (Principal G-bundle)
Let M be a smooth manifold and let G be a Lie group. A principal G-bundle over M is a smooth manifold P together with a smooth surjective map $\pi\colon P \to M$ and a right action of G on P satisfying equivariance and local triviality conditions.

We write this compactly as

$$G \hookrightarrow P \xrightarrow{\pi} M.$$

Example 1.1.2 (Frame bundles)
Let $E \to M$ be a $\mathbb{K}$-vector bundle of rank k. The frame bundle $\mathrm{Fr}(E)$ is the bundle whose fiber over $x \in M$ consists of all ordered bases of E_x. It is naturally a principal $\mathrm{GL}(k, \mathbb{K})$-bundle.

$$\mathrm{Fr}(E) = \coprod_{x \in M} \mathrm{Fr}_x(E).$$

If E is a real Riemannian vector bundle, one obtains the orthogonal frame bundle $O(E)$. If E is a complex Hermitian bundle, one obtains the unitary frame bundle $U(E)$.

Definition 1.1.3 (Connection on a principal bundle)
Let G be a Lie group with Lie algebra $\mathfrak{g}$. A connection on a principal bundle $G \hookrightarrow P \to M$ is a $\mathfrak{g}$-valued one-form $\omega \in \Omega^1(P, \mathfrak{g})$ satisfying the standard equivariance and normalization conditions.

Proposition 1.1.4 (Transformation of local gauge potentials)
Let ω be a connection on a principal bundle $G \hookrightarrow P \to M$, and let $\{U_i\}_{i\in I}$ be a trivializing cover with local gauges. The local gauge potentials are related on overlaps by

$$\mathcal{A}_j = \mathrm{Ad}_{g_{ij}^{-1}} \circ \mathcal{A}_i + g_{ij}^* \Theta.$$

If G is a matrix Lie group, this becomes

$$\mathcal{A}_j = g_{ij}^{-1} \mathcal{A}_i g_{ij} + g_{ij}^{-1} \mathrm{d} g_{ij}.$$

Definition 1.1.5 (Curvature)
The curvature of a connection ω is the two-form

$$\Omega := \mathrm{d}^\omega \omega.$$

Equivalently, it satisfies Cartan's structure equation

$$\Omega = \mathrm{d}\omega + \frac{1}{2}[\omega, \omega].$$

1.1.2 Associated bundles and matter fields

Let $G \hookrightarrow P \to M$ be a principal bundle, V a vector space, and $\rho\colon G \to \mathrm{GL}(V)$ a representation. The associated bundle is

$$P \times_\rho V = P \times V / G.$$

Sections of this bundle can be interpreted as matter fields in the language of physics.

1.1.3 The space of connections

Proposition 1.1.6 (Connections form an affine space)
Let $\mathrm{Conn}(P)$ be the set of connections of a principal bundle $G \hookrightarrow P \to M$. Then $\mathrm{Conn}(P)$ is an affine space modeled on the vector space of Ad-tensorial one-forms $\Omega^1_{\mathrm{Ad}}(P, \mathfrak{g})$.

1.2 Algebraic topology

The main algebraic-topological tools needed later are cohomology, characteristic classes, orientations, and the intersection form of four-manifolds.

1.2.1 Cohomology and cups

Definition 1.2.1 (Pairing of cohomology and homology)
The natural pairing of cohomology and homology is

$$\langle -, - \rangle \colon H^k(C, G) \otimes H_k(C) \to G,$$

given by evaluating a cohomology class on a homology class.

Theorem 1.2.2 (Universal coefficients for cohomology)
Let $C_\bullet$ be a chain complex of free abelian groups and let G be an abelian group. Then there is a split exact sequence

$$0 \to \mathrm{Ext}(H_{k-1}(C), G) \to H^k(C, G) \to \mathrm{Hom}(H_k(C), G) \to 0.$$

1.2.2 Orientations and caps

A local orientation at a point of an n-manifold is a choice of generator of the local homology group. A global orientation is a locally consistent family of such choices.

1.2.3 The intersection pairing

The intersection pairing on a closed oriented four-manifold is one of the central objects in this story. It may be defined using Poincaré duality and the cup product.

1.3 Characteristic classes

Characteristic classes measure obstructions to trivializing bundles and will later be used to formulate the existence of spin and Spin^c structures.

1.3.1 Classifying spaces for line bundles and the first Chern class

Complex line bundles are classified by maps into $\mathbb{CP}^\infty$, and the first Chern class records the corresponding cohomology class.

1.3.2 Čech cohomology and the first Stiefel-Whitney class

The first Stiefel-Whitney class detects the obstruction to orientability.

1.4 Hodge theory

This section would collect the Hodge-theoretic facts used later: harmonic representatives, Hodge decomposition, and self-dual forms.

1.5 Elliptic regularity

1.5.1 Sobolev spaces

Sobolev spaces provide the functional-analytic setting for elliptic differential operators.

1.5.2 Elliptic operators

Elliptic regularity is one of the analytic mechanisms behind smoothness statements for moduli spaces.

References

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[2] M. Sensoy, L. Kaplan, and M. Kandemir, "Evidential Deep Learning to Quantify Classification Uncertainty," in *Advances in Neural Information Processing Systems*, vol. 31, Curran Associates, Inc., 2018. Accessed: May 29, 2026. [Online]. Available: https://papers.nips.cc/paper_files/paper/2018/hash/a981f2b708044d6fb4a71a1463242520-Abstract.html